Abdul Monaf Chowdhury

University of Dhaka, Bangladesh

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Education

Master of Science in Robotics and Mechatronics Engineering July 2025 — Dec 2026
University of Dhaka, Bangladesh
Thesis: Language Guided Embodied Agents for Robotic Manipulation
Supervisor: Dr. Md Mehedi Hasan

Bachelor of Science in Robotics and Mechatronics Engineering Jan 2019 — Jan 2024
University of Dhaka, Bangladesh
*CGPA: **3.87/4.00**, Ranked* 2^{nd} ***place***

Relevant Coursework:
Artificial Intelligence, Introduction to Machine Learning, Digital Image Processing and Robot Vision, Digital Signal Processing, Human-Robot Interaction, Advanced Robotics

Skills:
Software: C, C++, Python, MATLAB, PyTorch, JAX, Flax, TensorFlow, LaTeX
Language: Fluent in both English and Bangla

Research Interests

Embodied AI, Multi-modal Learning, Computer Vision, Reinforcement Learning, Vision Language Models

Research Experience

Research Assistant Oct 2025 – Present
AVIS Lab, University of Dhaka *Dhaka, Bangladesh*
Supervisor: Dr. Md Mehedi Hasan

- Developing VLM guided embodied agents for robotic manipulation tasks, integrating multimodal perception with RL-based policy optimization
- Designing language-conditioned reflection mechanisms to improve task understanding and reward shaping, and evaluating the framework across simulated manipulation environments.

Research Assistant Feb 2024 – Sep 2025
MAIM Lab, University of Dhaka *Dhaka, Bangladesh*
Funding: Wellcome Leap (In Utero, California, USA)
PI: Dr. Niamh Nowlan, Co-PI: Dr. Abhishek Kumar Ghosh

- Collaborated with **University College Dublin** on the **Wellcome Leap In Utero** funded project titled “Translation of a Wearable Fetal Movement Monitor towards Stillbirth Prevention”
- Designed and implemented deep learning-based frameworks to analyze multimodal sensor signals from wearable belts to detect body movements, fetal kicks, and fetal hiccups
- Optimized signal processing of sensor data, presented hardware design feedback based on analytical findings, and validated hardware design changes
- Assisted in attaining **$1M** funding extension, and eventually helped secure translational funding from Wellcome Leap to launch a **startup**

Research Assistant Jan 2023 – Jan 2024
AVIS Lab, University of Dhaka *Dhaka, Bangladesh*
Supervisor: Dr. Md Mehedi Hasan; [Project Report]

- Collaborated on Unmanned Aerial Vehicle (UAV) based suspicious human activity recognition and drone surveillance
- Designed a hybrid model combining modified 3D CNN and FFT-based action recognition module for drone surveillance applications
- Built a lightweight deep learning pipeline using MobileNetV2 + BiLSTM for edge-based human activity detection, significantly reducing inference time

Professional Experience

Lecturer Mar 2026 – Apr 2026
Daffodil International University *Dhaka, Bangladesh*

- Instructed undergraduate junior and senior year students on Introduction to AI, and Control Systems courses, and took relevant labs.

Instructor May 2025 – Aug 2025
National Camp, Bangladesh AI Olympiad *Dhaka, Bangladesh*

- Instructed national camp students on Unsupervised Learning, Deep Learning, and Computer Vision algorithms and architectures, took relevant labs, and illustrated Deep Learning evaluation strategies and techniques

Publications

- *LaGEA: Language Guided Embodied Agents for Robotic Manipulation* **[Paper] [Code]**
  Abdul Monaf Chowdhury, AKM Moshiur Rahman Mazumder, Rabeya Akter Fariya, Safaeid Hossain
  43rd International Conference on Machine Learning, ICML '26

- *T3Time: Tri-Modal Time Series Forecasting via Adaptive Multi-Head Alignment and Residual Fusion* **[Paper] [Code]**
  Abdul Monaf Chowdhury, Rabeya Akter Fariya, Safaeid Hossain
  40th Annual AAAI Conference on Artificial Intelligence, AAAI '26

- *U-ActionNet: Dual-Pathway Fourier Networks with Region-of-Interest Module for Efficient Action Recognition in UAV Surveillance* **[Paper]**
  Abdul Monaf Chowdhury, Ahsan Imran, Md Mehedi Hasan, Riad Ahmed, AKM Azad, Salem A. Alyami
  IEEE Access, 2024. IF - 3.4

- *FFT-UAVNet: FFT Based Human Action Recognition for Drone Surveillance System* **[Paper]**
  Abdul Monaf Chowdhury, Ahsan Imran, Md Mehedi Hasan
  5th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI), 2023

Manuscript Submitted

- *Counting Through Occlusion: Framework for Open World Amodal Counting* **[Paper] [Code]**
  Safaeid Hossain, Rabeya Akter Fariya, **Abdul Monaf Chowdhury**, Md Mehedi Hasan, Md. Jubair Ahmed Sourov
  Winter Conference on Applications of Computer Vision - WACV 2027

Research Project

Open World Amodal Counting[Code] [Paper] Aug 2025 – Nov 2025
Multimodal Learning, Vision Language Model, Representation Learning

- Architected a hierarchical Feature Reconstruction Module that fuses visible spatial context with vision–language priors to reconstruct class-discriminative features at occluded locations, enabling amodal counts from RGB images
- Designed a Visual-Equivalence consistency objective with teacher–student alignment, anchoring reconstructions to unoccluded features and restoring robustness without depth sensors or structured layouts
- Built occlusion-augmented FSC-147 and CARPK benchmarks; achieved SOTA results under occlusion where MAE reduced by **32.2%↓** (FSC-147), **40.7%↓** (CARPK), **40.0%↓** (CAPTURe-Real), demonstrating strong cross-dataset generalization

Language Guided Embodied Agents[Code] [Paper] Mar 2025 – Sep 2025
Embodied AI, Vision Language Model, Reinforcement Learning

- Designed and integrated a **Qwen 2.5VL-3B** VLM driven "episodic reflection" module, automatically generating rich, natural-language self-assessments of each trial, highlighting successes and pinpointing failure causes to provide the agent with human-like introspection
- Fused multimodal reward signals by combining these verbal reflections with CLIP-style vision–language feedback from task descriptions and goal images, crafting a dense, semantically grounded reward model
- Engineered a reward-aligned **Soft Actor Critic**-based learning pipeline, where the enriched feedback loop accelerated exploration and policy refinement, consistently reducing training time and reliably converging on optimal behaviours across the Meta-World manipulation tasks

Tri-Modal Time Series Forecasting [Code] [Paper] Apr 2025 – Aug 2025

Large Language Model, Deep Learning, Signal Processing

- Architected an Adaptive Dynamic Multi-Head Cross-Modal Attention module with channel-wise residual skip-connections, enabling fine-grained alignment between temporal and auxiliary features and boosting representational capacity across modalities
- Engineered an **FFT**-based Frequency-Domain Processing pipeline, projecting real-valued spectra into learnable tokens and applying transformer-based attention with weighted pooling to extract robust spectral embeddings for each sensor channel
- Designed a Trainable Adaptive Rich-Horizon Gating Fusion to dynamically combine spectral and temporal encodings, and beat the state-of-the-art benchmark on multivariate time-series forecasting

Proxemics & Social Interaction Patterns in ASD Children Sep 2023 — Jan 2024

Human-Robot Interaction, Deep Learning

- Formulated a **YOLOv8**-based system to determine the ideal proxemics of autism spectrum disorder (ASD) children in front of **NAO** Robot
- Examined and analyzed the behavioural responses of twenty children diagnosed with ASD in the presence of specific actions performed by the NAO robot

Automatic Stock Trading [Report] Aug 2022 — Nov 2022

Reinforcement Learning

- Implemented Approximate Q Learning for three Bangladeshi stocks to generate Buy, Sell, and Hold orders
- Achieved 11% return of investment for the 3 stocks beating the DSE 30 index

Awards & Scholarships

- Dean's Award - for best Undergraduate Result, University of Dhaka, 2024
- Engineering Faculty Undergraduate Merit Scholarship, University of Dhaka, 2024
- 5th, Dataverse Challenge - ITVerse, Bangladesh, 2023; [Report]
- 2nd, Intra-Department Soccer Bot Championship, University of Dhaka, 2019
- Sylhet Board Scholarship, Higher Secondary Certificate Examination 2018

Workshop/Conference Attended

- 5th International Conference on Sustainable Technologies for Industry 5.0 (STI), Dhaka
- 40th Annual AAAI Conference on Artificial Intelligence, AAAI '26

Reviewer

- IEEE Access
- AAAI 26

Leadership/Volunteer Activities

General Secretary Mar 2022 — Feb 2024

RMEDU Student Club, University of Dhaka

Academic Team Mentor Sep 2019 — Aug 2022

Bangladesh Robot Olympiad

Program Co-Ordinator Jul 2021 -– Jun 2022

IEEE Robotics & Automation Society, University of Dhaka